UNIVERSITÀ DEGLI STUDI DI SALERNO
DIPARTIMENTO DI INFORMATICA
DIPARTIMENTO DI ECCELLENZA

CORSO DI LAUREA MAGISTRALE IN INFORMATICA

Enterprise Mobile Application Development

PROF.SSA RITA FRANCESE, DOTT. SABATO NOCERA

a.a. 2025-2026

Enterprise Application

R. Francese

“Enterprise Social Networking Helps Sanofi Pasteur Innovate and Improve Quality”

- Sanofi Pasteur is the vaccines division of the multinational pharmaceutical company Sanofi and the largest company in the world devoted entirely to vaccines. It is headquartered in Lyon, France, has nearly 15,000 employees worldwide, and produces more than 1 billion doses of vaccine per year to inoculate more than 500 million people around the globe. Sanofi Pasteur’s corporate vision is to work toward a world where no one suffers or dies from a vaccine-preventable disease. Every day the company invests more than € 1 million in research and development. Collaboration, sharing information, ongoing innovation, and rigorous pursuit of quality are essential for Sanofi Pasteur’s business success and commitment to improving the health of the world’s population.

- Until recently, the company lacked appropriate tools to encourage staff to have dialogues, share ideas, and work with other members of the company, including people that they might not know. As a large, centralized firm with a traditional hierarchical culture, initiatives were primarily driven from the top down. The company wanted to give employees more opportunities to experiment and innovate on their own, and adopted Microsoft Yammer as the platform for this change. Ideas for improvement can come from anywhere in the organization and through Yammer can be shared everywhere.

- Microsoft Yammer is an enterprise social networking platform for internal business uses, although it can also create external networks linking to suppliers, customers, and others outside the organization. Yammer enables employees to create groups to collaborate on projects and share and edit documents, and includes a news feed to find out what's happening within the company. A People Directory provides a searchable database of contact information, skills, and expertise. Yammer can be accessed through the web using desktop and mobile devices, and can be integrated with other Microsoft tools such as SharePoint and Office 365, to make other applications more "social." (SharePoint is Microsoft's platform for collaboration, document sharing, and document management. Office 365 is Microsoft's online service for its desktop productivity applications such as word processing, spreadsheet, electronic presentations, and data management.)
- How has Sanofi Pasteur benefited from becoming more "social"? Employees are using Yammer to share updates, ask for feedback, and connect volunteers to improvement initiatives. A recent project involving Yammer resulted in a 60 percent simplification of a key quality process at one manufacturing site, saving the company thousands of Euros, and reducing overall end-to-end process time. Through Yammer, employees spread the word about this improvement to other locations around the globe.

- Using Yammer, Sanofi employees set up activist networks for change in large manufacturing sites. Each group has attracted more than 1,000 people. These networks help create a more collegial, personal culture that helps people feel comfortable about making suggestions for improvements and working with other groups across the globe. They also provide management with observations about policies and procedures across departments and hierarchies that can be used to redesign the firm's manufacturing and business processes to increase quality and cost-effectiveness. For example, **a building operator shared his ideas about how to reduce waste when managing a specific material in his production facility. The new procedure for handling the material saved his facility more than 100,000 Euros per year and becam**e a global best practice at all Sanofi Pasteur production sites. Yammer-powered communities raised awareness of health, safety, and attention to detail, and more attention to these issues helped reduce human errors by 91 percent.

Enterprise Social Networking Helps Sanofi Pasteur Innovate and Improve Quality (1 of 2)

- Problem
  - Hierarchical top-down processes
  - Large geographically dispersed workforce
  - Lack of collaboration and idea sharing
- Solutions
  - Develop knowledge sharing strategy and goals
  - Redesign knowledge and collaboration processes
  - Change organizational culture
  - Implement Microsoft Yammer collaboration software

Enterprise Social Networking Helps Sanofi Pasteur Innovate and Improve Quality (2 of 2)

- Use of new technology to engage employees and enabled knowledge gathering and sharing
- Demonstrates how outdated processes can affect knowledge sharing and innovation
- Illustrates why organizations rely on information systems to improve performance and remain competitive

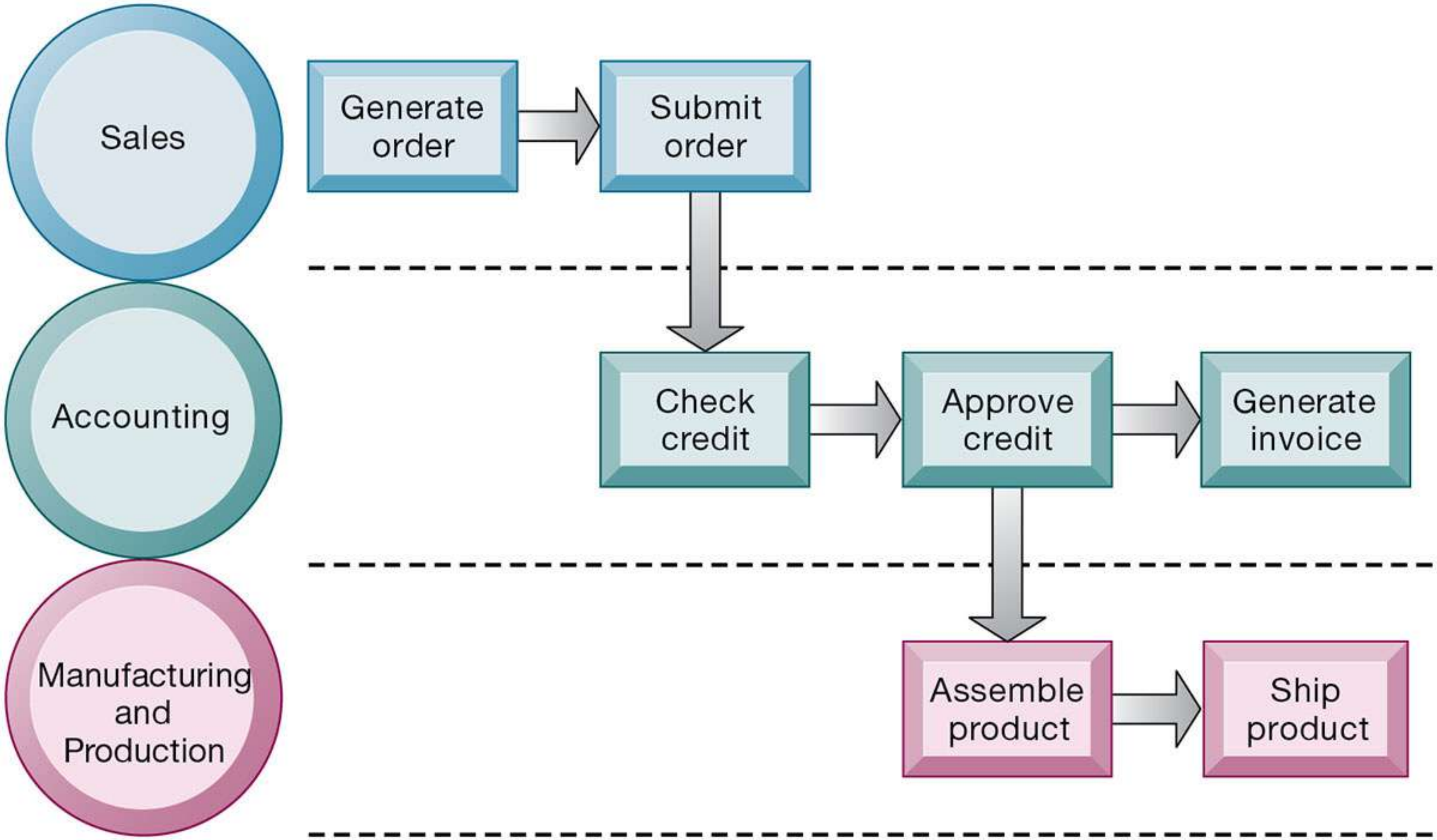
Business Processes (1 of 2)

- Business processes
  - Flows of material, information, knowledge
  - Logically related set of tasks that define how specific business tasks are performed
  - May be tied to functional area or be cross-functional
- Businesses: Can be seen as collection of business processes
- Business processes may be assets or liabilities

Business Processes (2 of 2)

- Examples of functional business processes
  - Manufacturing and production
    - Assembling the product
  - Sales and marketing
    - Identifying customers
  - Finance and accounting
    - Creating financial statements
  - Human resources
    - Hiring employees

Figure 2.1 The Order Fulfillment Process



Pearson

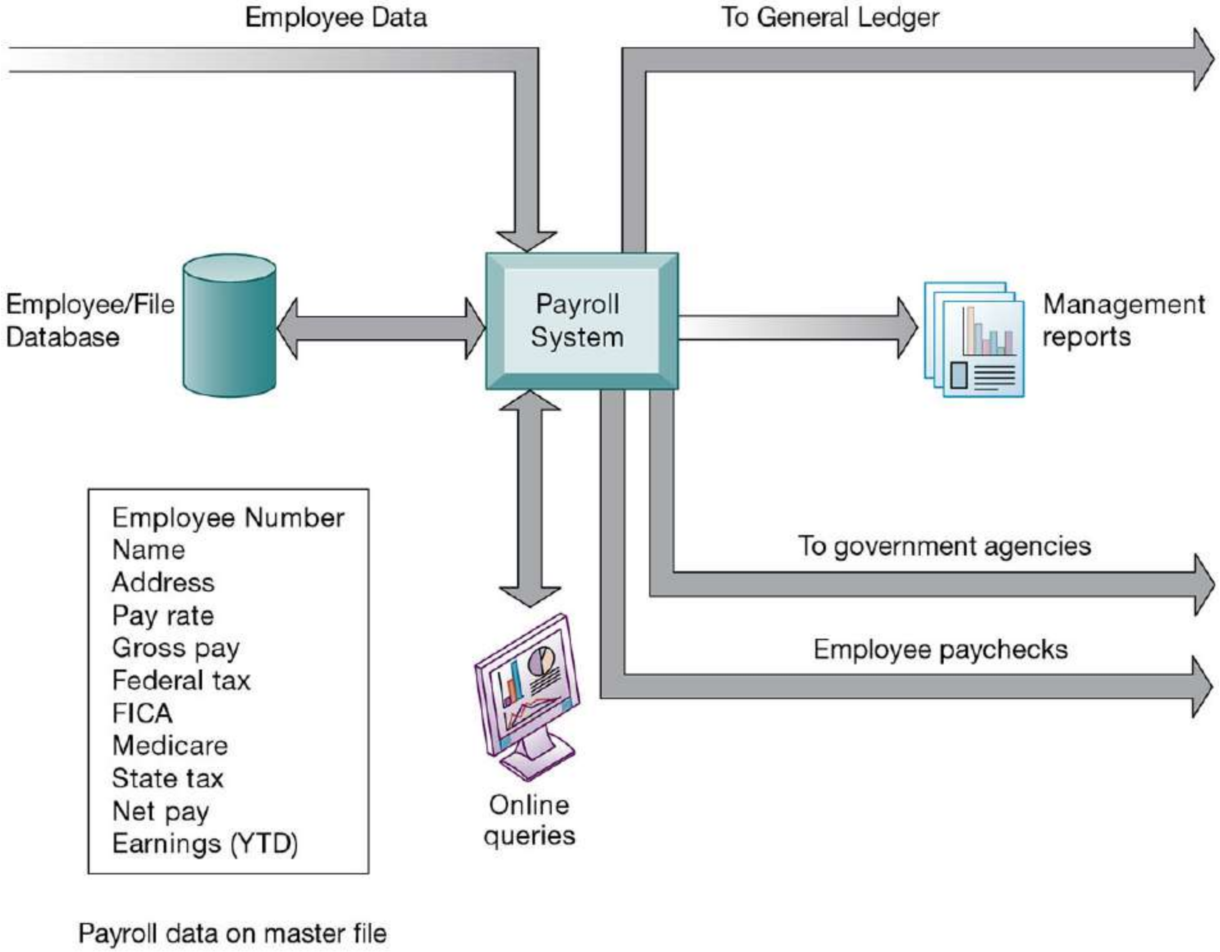
How Information Technology Improves Business Processes

- Increasing efficiency of existing processes
  - Automating steps that were manual
- Enabling entirely new processes
  - Changing flow of information
  - Replacing sequential steps with parallel steps
  - Eliminating delays in decision making
  - Supporting new business models

Systems for Different Management Groups (1 of 2)

- Transaction processing systems
  - Serve operational managers and staff
  - Perform and record daily routine transactions necessary to conduct business
    - Examples: sales order entry, payroll, shipping
  - Allow managers to monitor status of operations and relations with external environment
  - Serve predefined, structured goals and decision making

Figure 2.2 A Payroll TPS



Systems for Different Management Groups (2 of 2)

- Systems for business intelligence
  - Data and software tools for organizing and analyzing data
  - Used to help managers and users make improved decisions
- Management information systems
- Decision support systems
- Executive support systems

Management Information Systems

- Serve middle management
- Provide reports on firm’s current performance, based on data from TPS
- Provide answers to routine questions with predefined procedure for answering them
- Typically have little analytic capability

Figure 2.3 How Management Information Systems Obtain Their Data from the Organization’s TPS

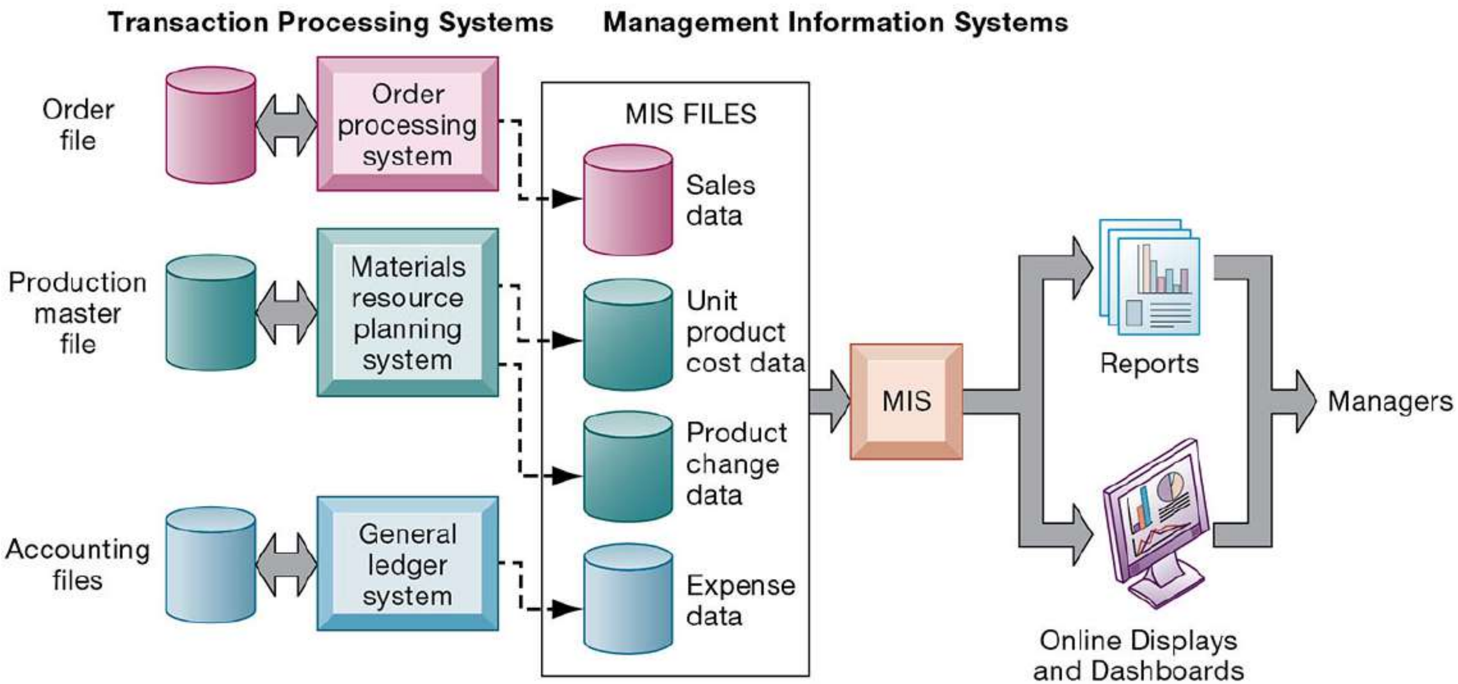


Figure 2.4 Sample MIS Report

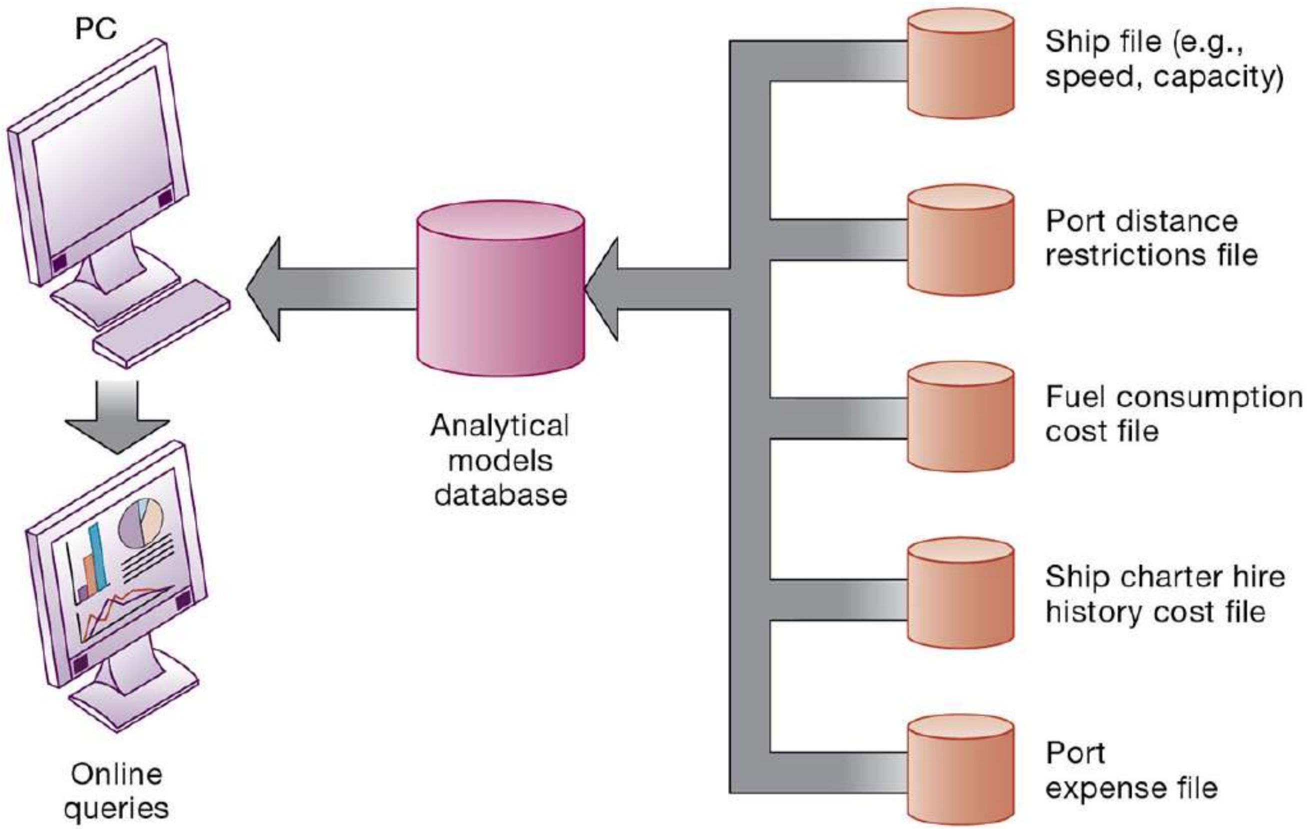
Consolidated Consumer Products Corporation Sales by Product and Sales Region: 2019

Product Code	Product Description	Sales Region	Actual Sales	Planned	Actual Versus Planned
4469	Carpet Cleaner	Northeast South Midwest West	4,066,700 3,778,112 4,867,001 4,003,440	4,800,000 3,750,000 4,600,000 4,400,000	0.85 1.01 1.06 0.91
	Total		16,715,253	17,550,000	0.95
5674	Room Freshener	Northeast South Midwest West	3,676,700 5,608,112 4,711,001 4,563,440	3,900,000 4,700,000 4,200,000 4,900,000	0.94 1.19 1.12 0.93
	Total		18,559,253	17,700,000	1.05

Decision Support Systems

- Serve middle management
- Support nonroutine decision making
  - Example: What is the impact on production schedule if December sales doubled?
- May use external information as well TPS / MIS data
- Model driven DSS
  - Voyage-estimating systems
- Data driven DSS
  - Intrawest's marketing analysis systems

Figure 2.5 Voyage-Estimating Decision-Support System



Executive Support Systems

- Support senior management
- Address nonroutine decisions
  - Requiring judgment, evaluation, and insight
- Incorporate data about external events (e.g., new tax laws or competitors) as well as summarized information from internal MIS and DSS
- Example: Digital dashboard with real-time view of firm's financial performance

Data Changes How NFL Teams Play the Game and How Fans See It

- All professional sports teams today collect detailed data on player and team performance, fan behavior, and sales, and increasingly use these data to drive decisions about every aspect of the business—marketing, ticketing, player evaluation, and TV and digital media deals. This includes the National Football League (NFL), which is increasingly turning to data to improve how its players and teams perform and how fans experience the game.

- Since 2014 the NFL has been capturing **player movement data** on the field by putting nickel-sized radio frequency identification (RFID) tags beneath players' shoulder pads to track every move they make. The information the sensors gather is used by NFL teams to improve their training and strategy, by commentators on live game broadcasts, and by fans attending games or using the NFL app on the Xbox One.

- The NFL's player tracking system is based on the Zebra Sports Solution developed by Zebra Technologies, a Chicago-based firm specializing in tracking technology that includes the bar codes on groceries and other consumer goods and radio frequency identification (RFID) technology.

- The Zebra Sports Solution system records players' speed, direction, location on the field, how far they ran on a play, and how long they were sprinting, jogging, or walking. The system can also determine what formation a team was in and how players' speed or acceleration affects their on-field performance. Want to know how hard Eli Manning is throwing passes or the force with which a ball arrives in the hands of receiver Odell Beckham? The system knows how to do all that.

- NFL players have RFID chips in their left and right shoulder pads that transmit data to 20 radio receivers strategically located in the lower and upper levels of stadiums to collect data about how each player moves, using metrics such as velocity, speed in miles per hour, and distance traveled. From there the data are transmitted to an on-site server computer, where Zebra's software matches an RFID tag to the correct player or official. The football also has a sensor transmitting location data. The data are generated in real-time as the game is being played. Each sensor transmits its location about 25 times per player.

- It takes just two seconds for data to be received by the motion sensors, analyzed, and pushed out to remote cloud computers run by Amazon Web Services for the NFL. From the NFL cloud computers, the data are shared with fans, broadcasters, and NFL teams. The data captured by the NFL are displayed to fans using the NFL Next Gen Stats website, NFL social media channels, and the NFL app on Windows 10 and the Xbox One. The data are also transmitted to the giant display screens in the arena to show fans during the game.

- The data have multiple uses. NFL teams use them to evaluate player and team performance and to analyze tactics, such as whether it might be better to press forward or to punt in a particular fourth-down situation. Data transmitted to broadcasters, to stadium screens, to Next Gen Stats, and to the Next Gen Stats feature of Microsoft's Xbox One NFL app help create a deeper fan experience that gets fans more involved in the game.

- Some of the statistics fans can now see on Next Gen Stats include Fastest Ball Carriers, Longest Tackles, Longest Plays, Passing Leaders, Rushing Leaders, and Receiving Leaders. Next Gen Stats also features charts for individual players and videos that explain the differences and similarities between players, teams, and games based on the data.

- While the data may be **entertaining f**or fans, they could prove **strategic** for the teams. Data markers for each play are recorded, including type of offense, type of defense, whether there was a huddle, all movement during the play, and the yard line where the ball was stopped. The NFL runs custom-created analytics to deliver visualizations of the data to each team within 24 hours of the game, via a custom-built web portal. The system displays charts and graphs as well as tabular data to let teams have more insight. Each NFL team may also hire its own data analyst to wring even more value from the data. The data are giving NFL fans, teams, coaches, and players a deeper look into the game they love.

Interactive Session: Management : Data Changes How NFL Teams Play the Game and How Fans See It

- Class discussion
  - What kinds of systems are illustrated in this case study? Where do they obtain their data? What do they do with the data? Describe some of the inputs and outputs of these systems.
  - What business functions do these systems support? Explain your answer.
  - How do the data about teams and players captured by the NFL help NFL football teams and the NFL itself make better decisions? Give examples of two decisions that were improved by the systems described in this case.
  - How did using data help the NFL and its teams improve the way they run their business?

Enterprise Applications

- Systems for linking the enterprise
- Span functional areas
- Execute business processes across the firm
- Include all levels of management
- Four major applications
  - Enterprise systems
  - Supply chain management systems
  - Customer relationship management systems
  - Knowledge management systems

Figure 2.6 Enterprise Application Architecture

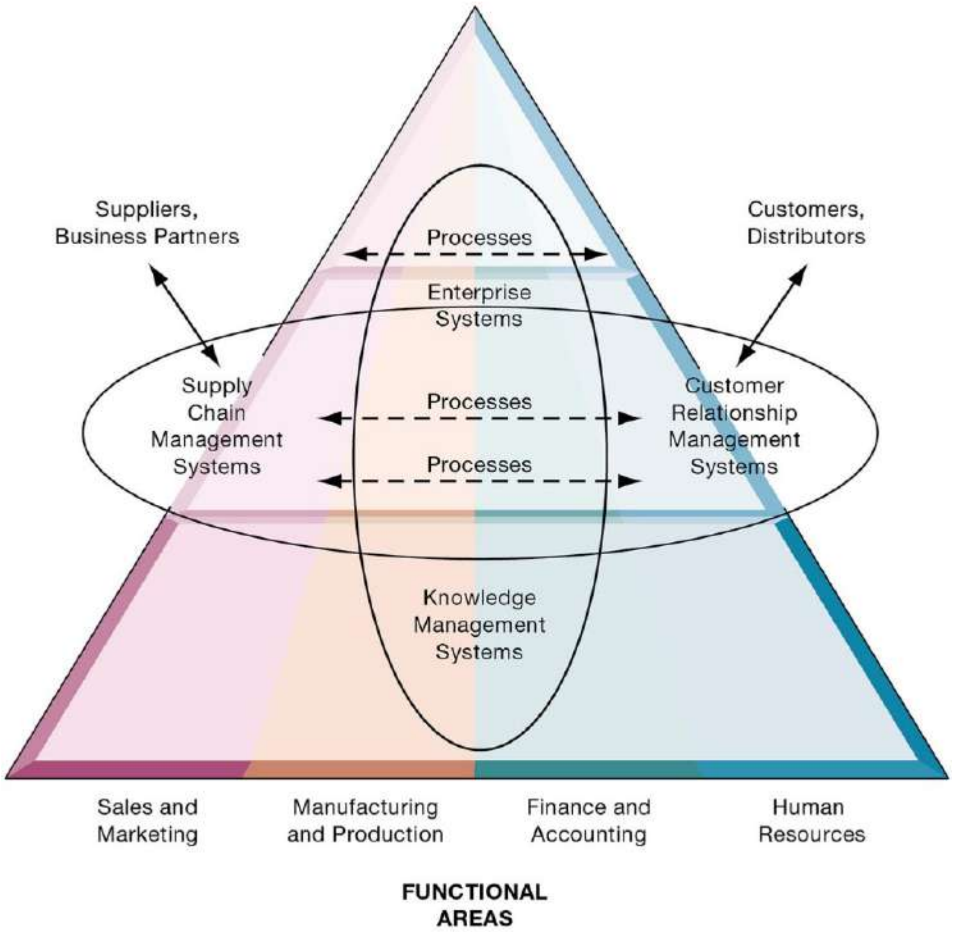
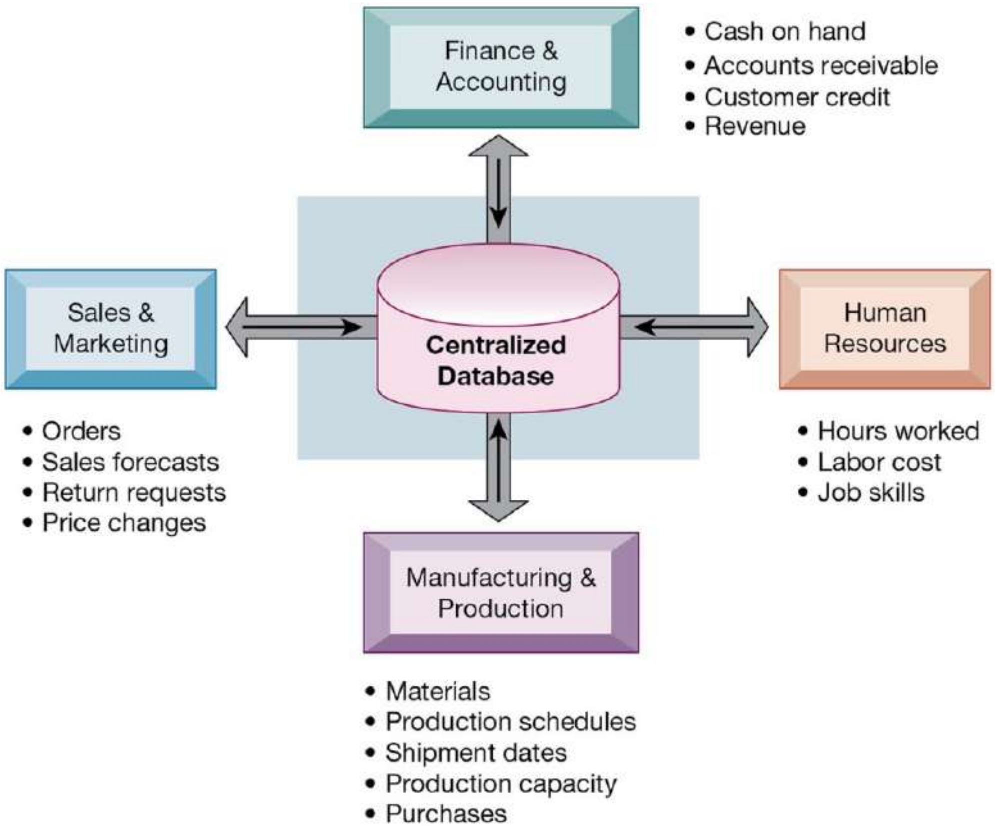


Figure 9.1 How Enterprise Systems Work



Enterprise Software

- Built around thousands of predefined business processes that reflect best practices
  - Finance and accounting
  - Human resources
  - Manufacturing and production
  - Sales and marketing
- To implement, firms:
  - Select functions of system they wish to use
  - Map business processes to software processes
    - Use software’s configuration tables for customizing

Business Value of Enterprise Systems

- Increase operational efficiency
- Provide firm-wide information to support decision making
- Enable rapid responses to customer requests for information or products
- Include analytical tools to evaluate overall organizational performance and improve decision-making

Enterprise Systems (ERP)

- Also called enterprise resource planning (ERP) systems
- Integrate data from key business processes into single system.
- Speed communication of information throughout firm.
- Enable greater flexibility in responding to customer requests, greater accuracy in order fulfillment.
- Enable managers to assemble overall view of operations.

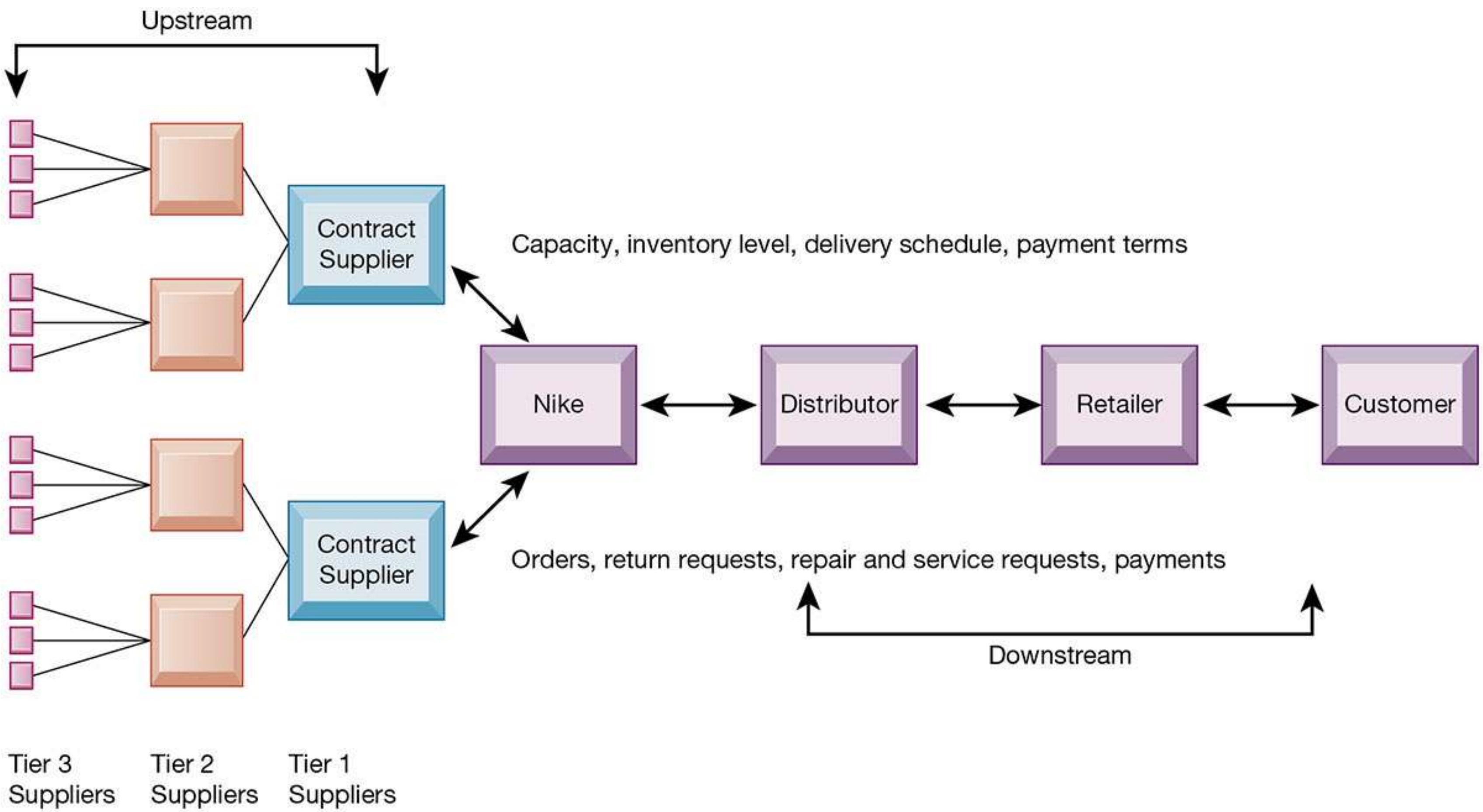
Supply Chain Management (SCM) Systems

- Manage relationships with suppliers, purchasing firms, distributors, and logistics companies.
- Manage shared information about orders, production, inventory levels, and so on.
- Goal is to move correct amount of product from source to point of consumption as quickly as possible and at lowest cost
- Type of interorganizational system: Automating flow of information across organizational boundaries

The Supply Chain

- Network of organizations and processes for:
  - Procuring materials
  - Transforming materials into products
  - Distributing the products
- Upstream supply chain
- Downstream supply chain
- Internal supply chain

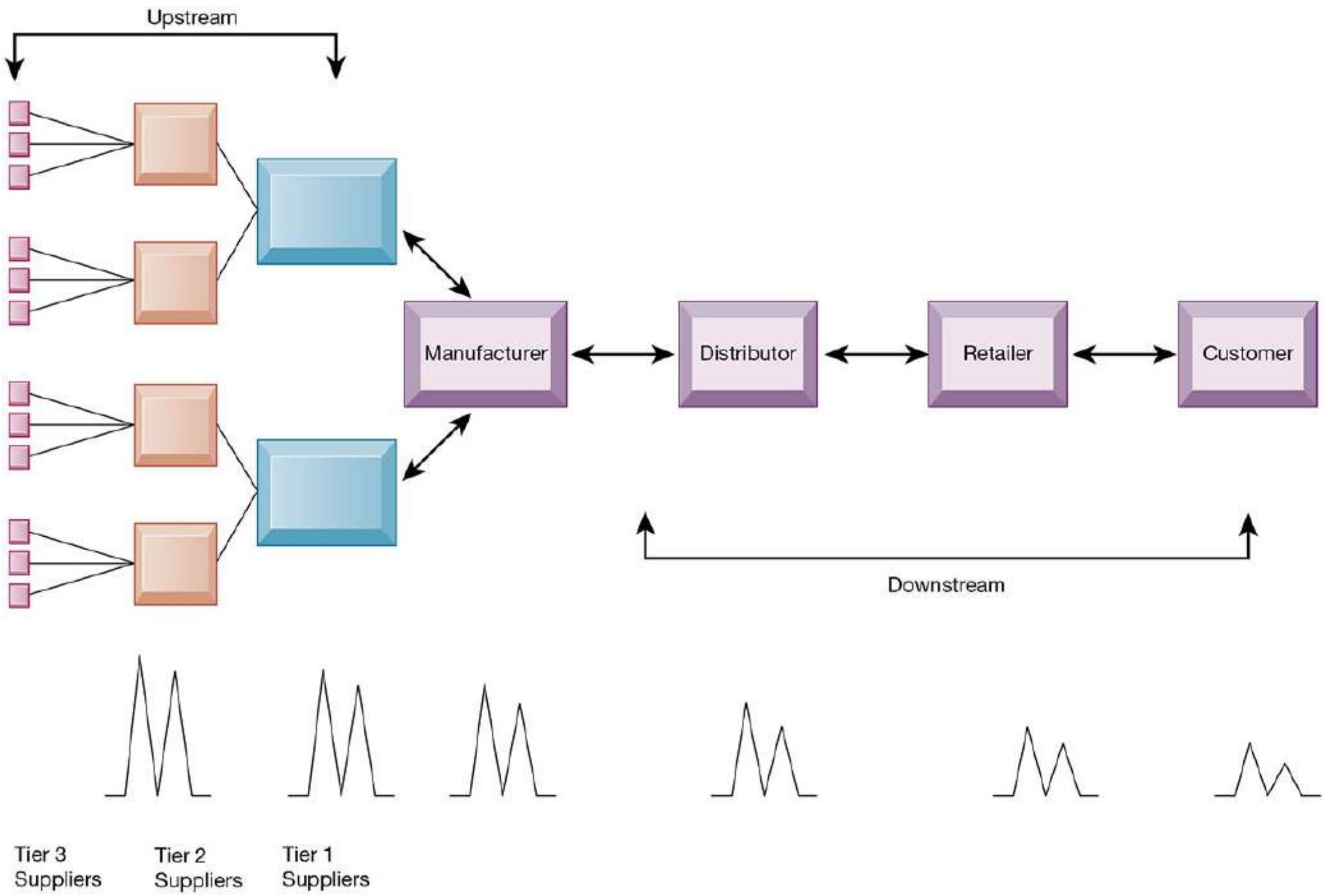
Figure 9.2 Nike's Supply Chain



Supply Chain Management

- Inefficiencies cut into a company's operating costs
  - Can waste up to 25 percent of operating expenses
- Just-in-time strategy
  - Components arrive as they are needed
  - Finished goods shipped after leaving assembly line
- Safety stock: buffer for lack of flexibility in supply chain
- Bullwhip effect
  - Information about product demand gets distorted as it passes from one entity to next across supply chain

Figure 9.3 The Bullwhip Effect



Supply Chain Management Software

- Supply chain planning systems
  - Model existing supply chain
  - Enable demand planning
  - Optimize sourcing, manufacturing plans
  - Establish inventory levels
  - Identify transportation modes
- Supply chain execution systems
  - Manage flow of products through distribution centers and warehouses

Global Supply Chains and the Internet

- Global supply chain issues
  - Greater geographical distances, time differences
  - Participants from different countries
    - Different performance standards
    - Different legal requirements
- Internet helps manage global complexities
  - Warehouse management
  - Transportation management
  - Logistics
  - Outsourcing

Demand-Driven Supply Chains: From Push to Pull Manufacturing and Efficient Customer Response

- Push-based model (build-to-stock)
  - Earlier SCM systems
  - Schedules based on best guesses of demand
- Pull-based model (demand-driven)
  - Web-based
  - Customer orders trigger events in supply chain
- Internet enables move from sequential supply chains to concurrent supply chains
  - Complex networks of suppliers can adjust immediately

Figure 9.4 Push- Versus Pull-Based Supply Chain Models

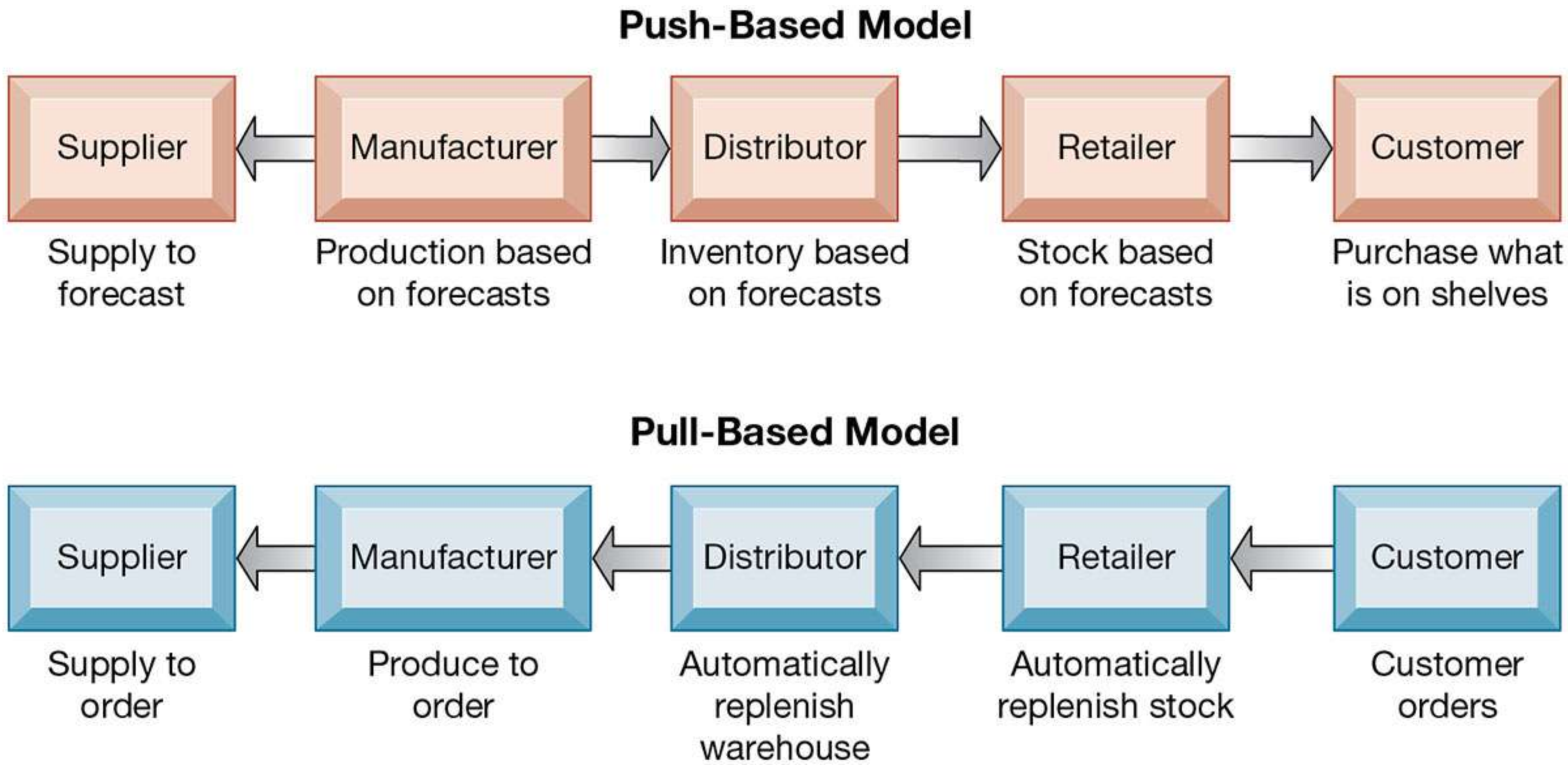
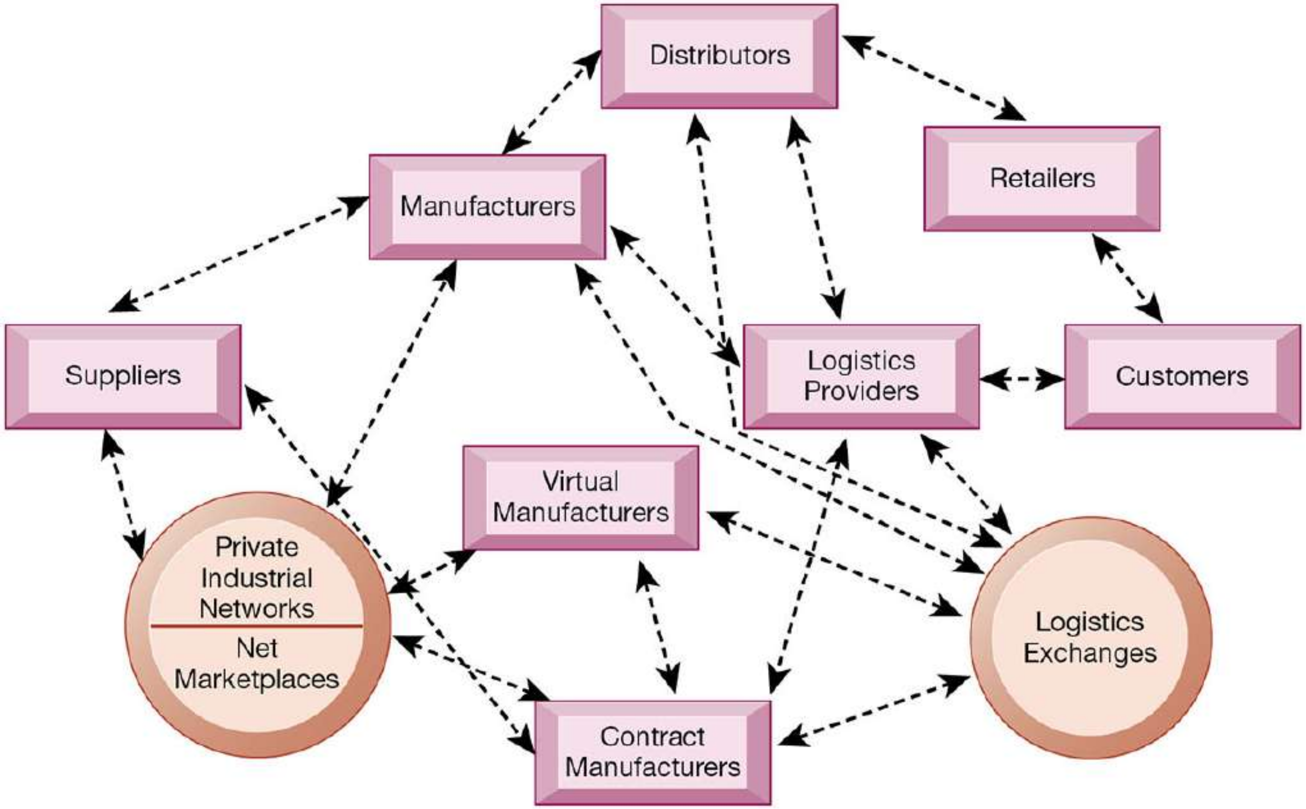


Figure 9.5 The Emerging Internet-Driven Supply Chain



Business Value of Supply Chain Management Systems

- Match supply to demand
- Reduce inventory levels
- Improve delivery service
- Speed product time to market
- Use assets more effectively
  - Total supply chain costs can be 75 percent of operating budget
- Increase sales

Interactive Session: Management: Soma Bay Prospers with ERP in the Cloud

- Class discussion
  - Identify and describe the problem discussed in this case. What management, organization, and technology factors contributed to the problem?
  - Why was an ERP system required for a solution? How did having a cloud-based ERP system contribute to the solution?
  - What were the business benefits of Soma Bay's new enterprise system? How did it change decision making and the way the company operated?

Customer Relationship Management (CRM) Systems

- Help manage relationship with customers.
- Coordinate business processes that deal with customers in sales, marketing, and customer service
- Goals:
  - Optimize revenue
  - Improve customer satisfaction
  - Increase customer retention
  - Identify and retain most profitable customers
  - Increase sales

Customer Relationship Management

- Knowing the customer
- In large businesses, too many customers and too many ways customers interact with firm
- CRM systems
  - Capture and integrate customer data from all over the organization
  - Consolidate and analyze customer data
  - Distribute customer information to various systems and customer touch points across enterprise
  - Provide single enterprise view of customers

Figure 9.6 Customer Relationship Management (CRM)

Customer Relationship Management Software (1 of 2)

- Packages range from niche tools to large-scale enterprise applications
- More comprehensive packages have modules for:
  - Partner relationship management (PRM)
    - Integrating lead generation, pricing, promotions, order configurations, and availability
    - Tools to assess partners' performances
  - Employee relationship management (ERM)
    - Setting objectives, employee performance management, performance-based compensation, employee training

Customer Relationship Management Software (2 of 2)

- CRM packages typically include tools for:
  - Sales force automation (SFA)
    - Sales prospect and contact information
    - Sales quote generation capabilities
  - Customer service
    - Assigning and managing customer service requests
    - Web-based self-service capabilities
  - Marketing
    - Capturing prospect and customer data, scheduling and tracking direct-marketing mailings or e-mail
    - Cross-selling

Figure 9.7 How CRM Systems Support Marketing

Responses by Channel for January 2019 Promotional Campaign

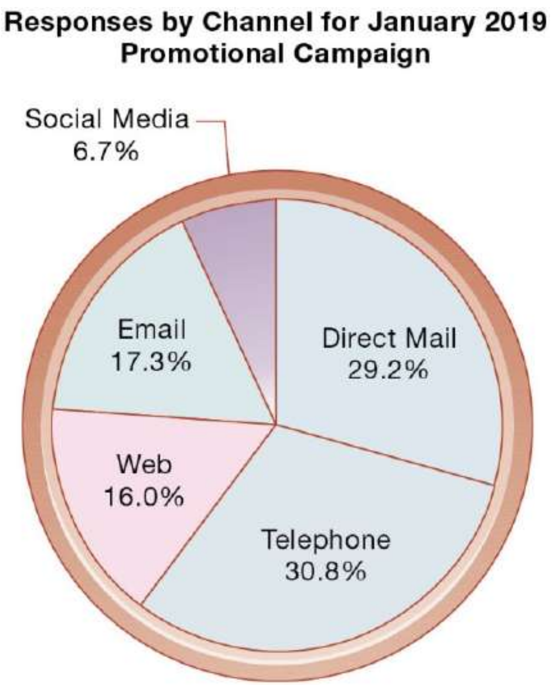
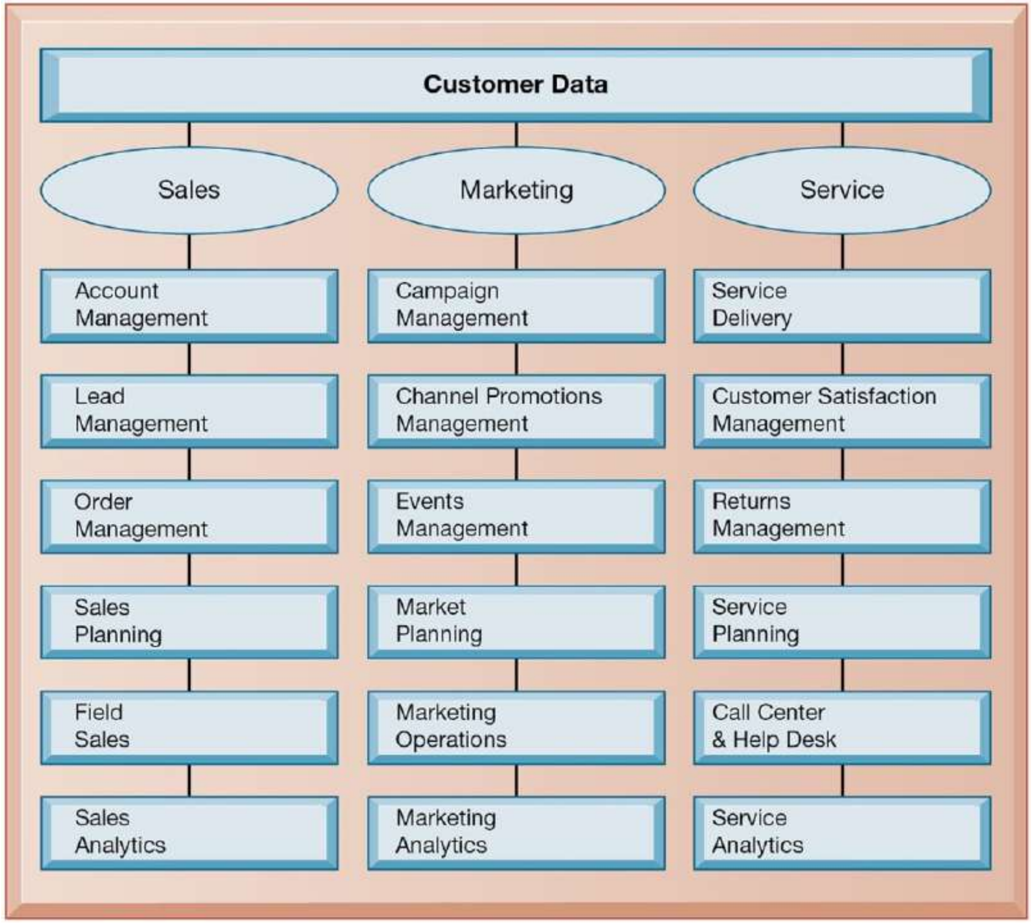
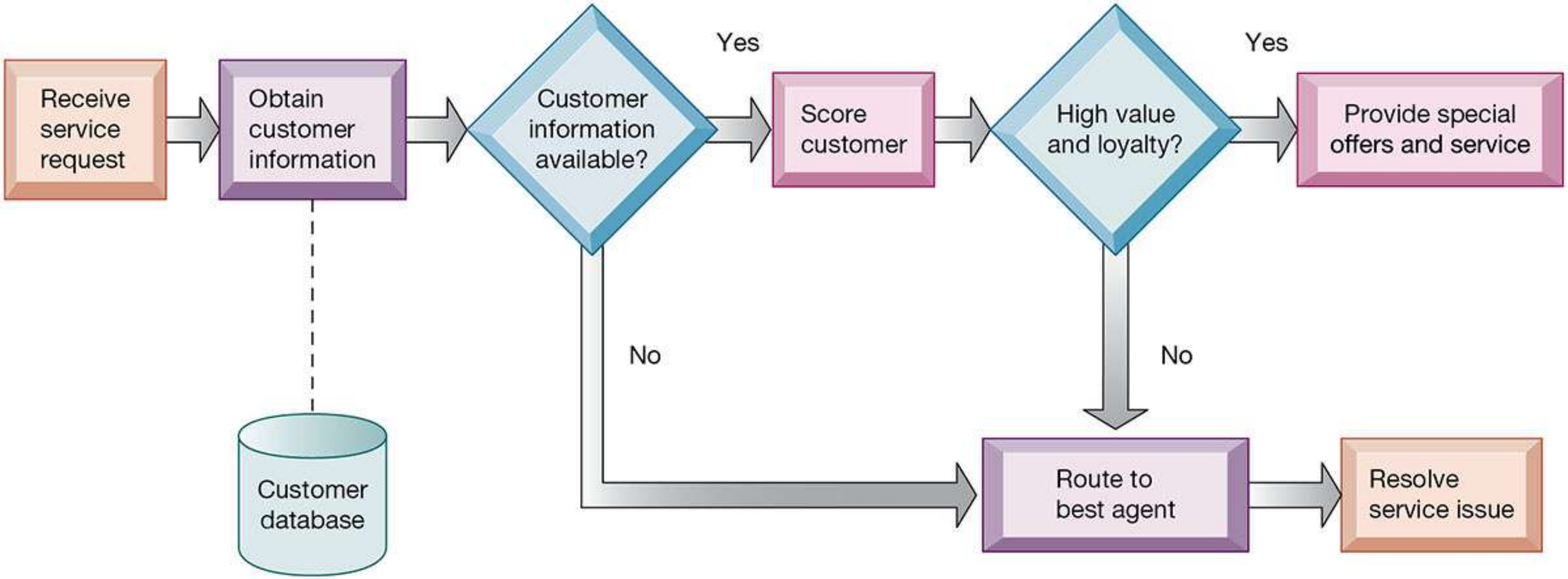


Figure 9.8 CRM Software Capabilities



Pearson

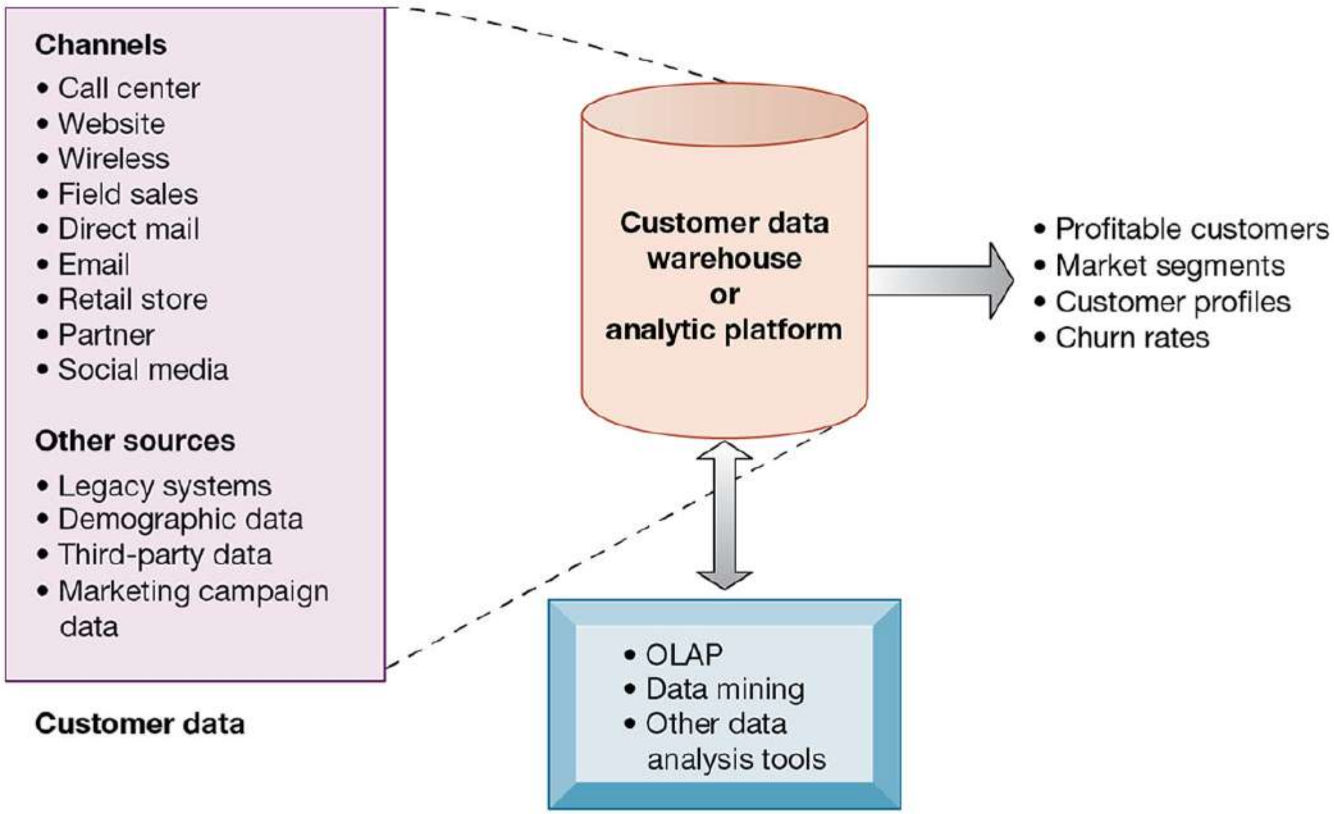
Figure 9.9 Customer Loyalty Management Process Map



Operational and Analytical CRM

- Operational CRM
  - Customer-facing applications
  - Sales force automation call center and customer service support
  - Marketing automation
- Analytical CRM
  - Based on data warehouses populated by operational CRM systems and customer touch points
  - Analyzes customer data (OLAP, data mining, etc.)
    - Customer lifetime value (CLTV)

Figure 9.10 Analytical CRM Data Warehouse



Interactive Session – Organizations: Kenya Airways Flies High with Customer Relationship Management

- Class discussion
  - What was the problem at Kenya Airways described in this case? What people, organization, and technology factors contributed to this problem?
  - What was the relationship of customer relationship management to Kenya Airway's business performance and business strategy?
  - Describe Kenya Airway's solution to its problem. What people, organization, and technology issues had to be addressed by the solution?
  - How effective was this solution? How did it affect the way Kenya Airways ran its business and its business performance?

Business Value of Customer Relationship Management Systems

- Business value of CRM systems
  - Increased customer satisfaction
  - Reduced direct-marketing costs
  - More effective marketing
  - Lower costs for customer acquisition/retention
  - Increased sales revenue
- Churn rate
  - Number of customers who stop using or purchasing products or services from a company
  - Indicator of growth or decline of firm’s customer base

Enterprise Application Challenges

- Expensive to purchase and implement enterprise applications
  - Multi-million dollar projects in 2018
  - Long development times
- Technology changes
- Business process changes
- Organizational learning, changes
- Switching costs, dependence on software vendors
- Data standardization, management, cleansing

Next-Generation Enterprise Applications (1 of 2)

- Enterprise solutions/suites
  - Make applications more flexible, web-enabled, integrated with other systems
- SOA standards
- Open-source applications
- On-demand solutions
- Cloud-based versions
- Functionality for mobile platform

Next-Generation Enterprise Applications (2 of 2)

- Social CRM
  - Incorporating social networking technologies
  - Company social networks
  - Monitor social media activity; social media analytics
  - Manage social and web-based campaigns
- Business intelligence
  - Inclusion of BI with enterprise applications
  - Flexible reporting, ad hoc analysis, "what-if" scenarios, digital dashboards, data visualization

Knowledge Management Systems (KMS)

- Manage processes for capturing and applying knowledge and expertise
- Collect relevant knowledge and make it available wherever needed in the enterprise to improve business processes and management decisions.
- Link firm to external sources of knowledge

Intranets and Extranets

- Technology platforms that increase integration and expedite the flow of information
- Intranets:
  - Internal networks based on Internet standards
  - Often are private access area in company’s Web site
- Extranets:
  - Company Web sites accessible only to authorized vendors and suppliers
  - Facilitate collaboration

E-Business, E-Commerce, and E-Government

- E-business
  - Use of digital technology and Internet to drive major business processes
- E-commerce
  - Subset of e-business
  - Buying and selling goods and services through Internet
- E-government
  - Using Internet technology to deliver information and services to citizens, employees, and businesses

What is Collaboration?

- Collaboration
  - Short lived or long term
  - Informal or formal (teams)
- Growing importance of collaboration
  - Changing nature of work
  - Growth of professional work—“interaction jobs”
  - Changing organization of the firm
  - Changing scope of the firm
  - Emphasis on innovation
  - Changing culture of work

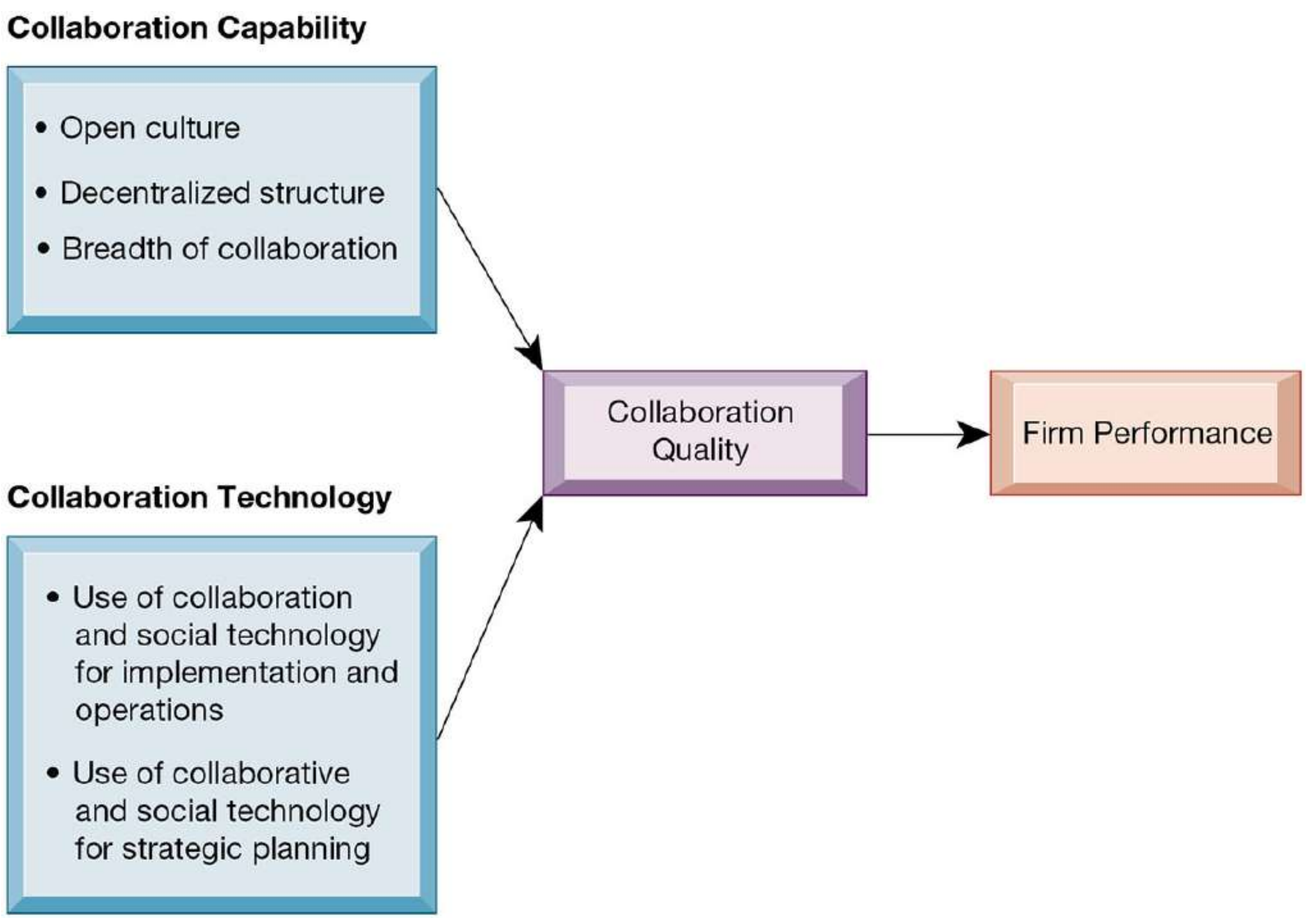
What is Social Business?

- Social business
  - Use of social networking platforms (internal and external) to engage employees, customers, and suppliers
- Aims to deepen interactions and expedite information sharing
- “Conversations” to strengthen bonds with customers
- Requires information transparency
- Seen as way to drive operational efficiency, spur innovation, accelerate decision making

Business Benefits of Collaboration and Teamwork

- Investment in collaboration technology can return large rewards, especially in sales and marketing, research and development
- Productivity: Sharing knowledge and resolving problems
- Quality: Faster resolution of quality issues
- Innovation: More ideas for products and services
- Customer service: Complaints handled more rapidly
- Financial performance: Generated by improvements in factors above

Figure 2.7 Requirements for Collaboration



Pearson

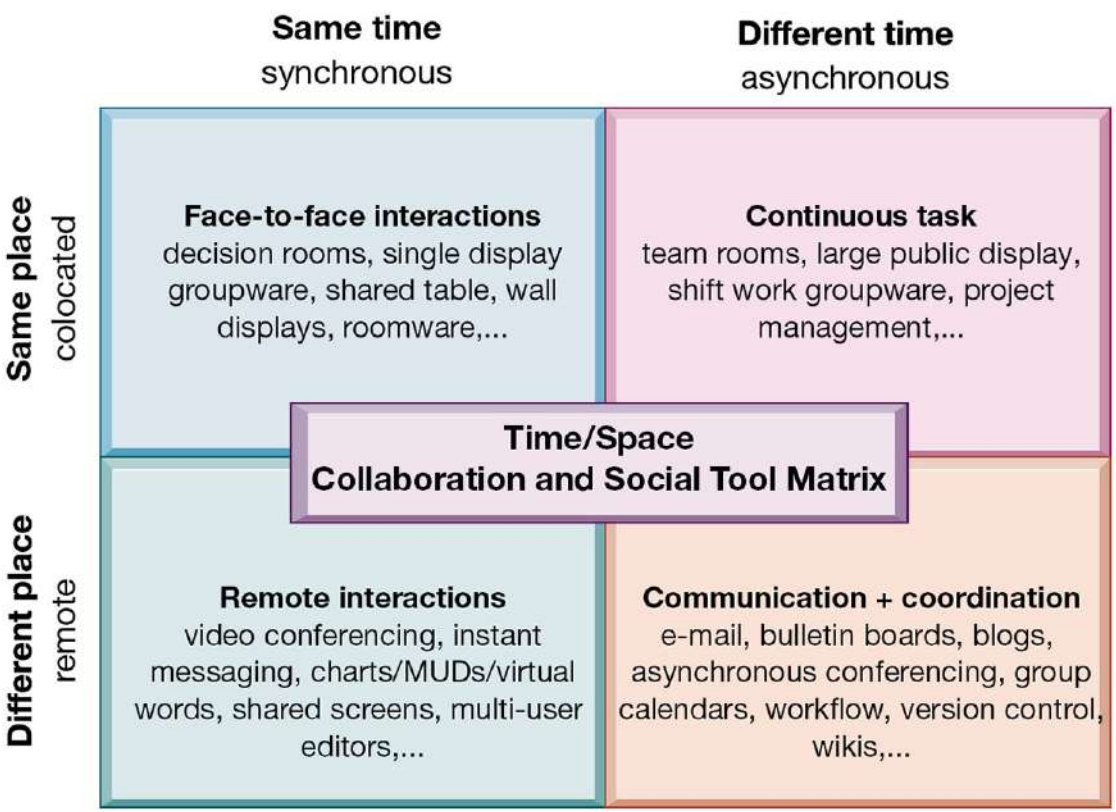
Building a Collaborative Culture and Business Processes

- “Command and control” organizations
  - No value placed on teamwork or lower-level participation in decisions
- Collaborative business culture
  - Senior managers rely on teams of employees
  - Policies, products, designs, processes, and systems rely on teams
  - The managers purpose is to build teams

Tools and Technologies for Collaboration and Social Business

- E-mail and instant messaging (IM)
- Wikis
- Virtual worlds
- Collaboration and social business platforms
  - Virtual meeting systems (telepresence)
  - Cloud collaboration services (Google Drive, Google Docs, etc.)
  - Microsoft SharePoint and IBM Notes
  - Enterprise social networking tools

Figure 2.8 The Time/Space Collaboration and Social Tool Matrix



The Information Systems Department

- Often headed by chief information officer (CIO)
  - Other senior positions include chief security officer (CSO), chief knowledge officer (CKO), chief privacy officer (CPO), chief data officer (CDO)
- Programmers
- Systems analysts
- Information systems managers
- End users

Organizing the Information Systems Function

- IT governance
  - Strategies and policies for using IT in the organization
  - Decision rights
  - Accountability
  - Organization of information systems function
    - Centralized, decentralized, and so on